Signals & Systems (151-0575-01) Exam - February 8th, 2022

Combined question sheet and scoring rubric generated from JSON source. Criterion headings match rubric_manifest.json.

q1 - Consider the causal continuous-time LTI system G_c with transfer func... - 1 point:

Question (q1): Consider the causal continuous-time LTI system G_c with transfer function H(s). Select the correct statement regarding discretization mapping via sigmoid function g(x).
Options:
(1) The resulting discretized system is always stable
(2) It is not possible to determine the stability of the system with the given information
(3) None of the other statements is correct
(4) If the continuous-time system is stable, then the discretized system is unstable
(5) If the continuous-time system is unstable, then the discretized system is unstable
(6) The resulting discretized system is always unstable
Award full credit only for unambiguous selection of option (1): The resulting discretized system is always stable. Otherwise 0 unless the rubric allows partial credit.

q2 - Consider the same discretization mapping. Select the correct statemen... - 1 point:

Question (q2): Consider the same discretization mapping. Select the correct statement regarding time-invariance.
Options:
(1) It is not possible to determine time-invariance of the system with the given information
(2) The resulting discretized system is always non-causal
(3) None of the other statements is correct
(4) The resulting discretized system is always nonlinear
(5) The resulting discretized system is always time-invariant
Award full credit only for unambiguous selection of option (5): The resulting discretized system is always time-invariant. Otherwise 0 unless the rubric allows partial credit.

q3 - Mark the true statement regarding Exact and Euler discretization as T... - 1 point:

Question (q3): Mark the true statement regarding Exact and Euler discretization as Ts -> 0.
Options:
(1) In Exact discretization, when Ts -> 0 then the input matrix Bd -> I
(2) In Euler discretization, when Ts -> 0 then the state matrix Ad -> 0
(3) None of the other statements is correct
(4) In both Euler and Exact discretization, when Ts -> 0 then the state matrix Ad -> I
Award full credit only for unambiguous selection of option (4): In both Euler and Exact discretization, when Ts -> 0 then the state matrix Ad -> I. Otherwise 0 unless the rubric allows partial credit.

q4 - Consider H(s) = 1 / (s - 0.5) discretized using Euler discretization... - 1 point:

Question (q4): Consider H(s) = 1 / (s - 0.5) discretized using Euler discretization with sampling time Ts. What is the ratio kr between the CT DC gain and the DT DC gain?
Options:
(1) kr = exp(-Ts)
(2) kr = -Ts
(3) kr = exp(Ts)
(4) kr = 1
(5) kr = 0
(6) kr = Ts
Award full credit only for unambiguous selection of option (4): kr = 1. Otherwise 0 unless the rubric allows partial credit.

q5 - For which sampling times Ts is the discretized system from Question 4... - 1 point:

Question (q5): For which sampling times Ts is the discretized system from Question 4 unstable?
Options:
(1) 0 <= Ts <= 0.5
(2) 0 <= Ts <= 2
(3) None of the other statements is correct

(4) Ts >= 1
(5) Ts >= 0
(6) Ts >= 2
(7) Ts >= 0.5
(8) 0 <= Ts <= 4
Award full credit only for unambiguous selection of option (5): Ts >= 0. Otherwise 0 unless the rubric allows partial credit.

q6 - What is the discrete-time frequency Omega_1 for the sampled signal x_... - 1 point:

Question (q6): What is the discrete-time frequency Omega_1 for the sampled signal x_1[n] = cos(Omega_1 n)?
Options:
(1) Omega_1 = pi/2
(2) Omega_1 = pi/25
(3) Omega_1 = 5pi/6
(4) Omega_1 = 2/3
(5) Omega_1 = 1/50
(6) Omega_1 = pi
Award full credit only for unambiguous selection of option (2): Omega_1 = pi/25. Otherwise 0 unless the rubric allows partial credit.

q7 - What is the period N_2 of the discrete-time signal x_2[n]? - 1 point:

Question (q7): What is the period N_2 of the discrete-time signal x_2[n]?
Options:
(1) N_2 = 60
(2) N_2 = 100
(3) N_2 = 20
(4) N_2 = 120
(5) N_2 = 70
(6) N_2 = 30
Award full credit only for unambiguous selection of option (1): N_2 = 60. Otherwise 0 unless the rubric allows partial credit.

q8 - What is the fundamental period N of the discrete-time signal x[n] = c... - 1 point:

Question (q8): What is the fundamental period N of the discrete-time signal x[n] = cos(pi/7 * n) + sin(pi/8 * n)?
Options:
(1) N = 30
(2) N = 56
(3) N = 84
(4) N = 132
(5) N = 112
(6) N = 24
Award full credit only for unambiguous selection of option (5): N = 112. Otherwise 0 unless the rubric allows partial credit.

q9 - Select the answer containing all linear systems among G_A, G_B, G_C,... - 1 point:

Question (q9): Select the answer containing all linear systems among G_A, G_B, G_C, G_D.
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) G_A, G_B
(6) G_A, G_C
(7) G_A, G_D
(8) G_B, G_C
(9) G_B, G_D
(10) G_C, G_D
(11) G_A, G_B, G_C
(12) G_A, G_B, G_D
(13) G_A, G_C, G_D
(14) G_B, G_C, G_D

(15) G_A, G_B, G_C, G_D
Award full credit only for unambiguous selection of option (15): G_A, G_B, G_C, G_D. Otherwise 0 unless the rubric allows partial credit.

q10 - Select the answer containing all time-invariant systems among G_A, G_... - 1 point:

Question (q10): Select the answer containing all time-invariant systems among G_A, G_B, G_C, G_D.
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) G_A, G_B
(6) G_A, G_C
(7) G_A, G_D
(8) G_B, G_C
(9) G_B, G_D
(10) G_C, G_D
(11) G_A, G_B, G_C
(12) G_A, G_B, G_D
(13) G_A, G_C, G_D
(14) G_B, G_C, G_D
(15) G_A, G_B, G_C, G_D
Award full credit only for unambiguous selection of option (11): G_A, G_B, G_C. Otherwise 0 unless the rubric allows partial credit.

q11 - Select the answer containing all causal systems among G_A, G_B, G_C,... - 1 point:

Question (q11): Select the answer containing all causal systems among G_A, G_B, G_C, G_D.
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) G_A, G_B
(6) G_A, G_C
(7) G_A, G_D
(8) G_B, G_C
(9) G_B, G_D
(10) G_C, G_D
(11) G_A, G_B, G_C
(12) G_A, G_B, G_D
(13) G_A, G_C, G_D
(14) G_B, G_C, G_D
(15) G_A, G_B, G_C, G_D
Award full credit only for unambiguous selection of option (5): G_A, G_B. Otherwise 0 unless the rubric allows partial credit.

q12 - Select the answer containing all BIBO LTI stable systems among G_A, G... - 1 point:

Question (q12): Select the answer containing all BIBO LTI stable systems among G_A, G_B, G_C, G_D.
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) G_A, G_B
(6) G_A, G_C
(7) G_A, G_D
(8) G_B, G_C
(9) G_B, G_D
(10) G_C, G_D
(11) G_A, G_B, G_C

(12) G_A, G_B, G_D
(13) G_A, G_C, G_D
(14) G_B, G_C, G_D
(15) G_A, G_B, G_C, G_D
Award full credit only for unambiguous selection of option (1): G_A. Otherwise 0 unless the rubric allows partial credit.

q13 - What is the impulse response of G given the step response relation an... - 1 point:

Question (q13): What is the impulse response of G given the step response relation and output observations?
Options:
(1) {h[n]} = {..., 0, 2, 1/4, 1/3, 1/4, 0, 0, ...}
(2) None of the other statements is correct
(3) {h[n]} = {..., 0, 2, 2, 0, ...}
(4) {h[n]} = {..., 0, 1/4, 2, 2, 2, 0, ...}
(5) {h[n]} = {..., 0, 1/4, 1, 0, ...}
(6) {h[n]} = {..., 0, 1/3, -2/3, 2/3, 0, 0, ...}
Award full credit only for unambiguous selection of option (6): {h[n]} = {..., 0, 1/3, -2/3, 2/3, 0, 0, ...}. Otherwise 0 unless the rubric allows partial credit.

q14 - Select the correct statement regarding system G from Question 13. - 1 point:

Question (q14): Select the correct statement regarding system G from Question 13.
Options:
(1) The system G is non-causal and unstable.
(2) None of the other statements is correct
(3) The system G is non-causal and stable.
(4) It is not possible to determine if the system G is stable.
(5) The system G is causal and stable.
(6) The system G is causal and unstable.
Award full credit only for unambiguous selection of option (5): The system G is causal and stable.. Otherwise 0 unless the rubric allows partial credit.

q15 - Let G be an LTI system with given step response measurements r[0]=1,... - 1 point:

Question (q15): Let G be an LTI system with given step response measurements r[0]=1, r[1]=2, r[2]=3.1. Select the correct statement.
Options:
(1) None of the other statements is correct
(2) The system G is unstable and causal
(3) The system G is stable and causal
(4) It is not possible to determine stability and causality with the given information
(5) The system G is stable and non-causal
(6) The system G is unstable and non-causal
Award full credit only for unambiguous selection of option (2): The system G is unstable and causal. Otherwise 0 unless the rubric allows partial credit.

q16 - Given that q[1]=0.5 for the system in Question 15, select the correct... - 1 point:

Question (q16): Given that q[1]=0.5 for the system in Question 15, select the correct statement.
Options:
(1) None of the other statements is correct
(2) The output matrix of the system G is c = 3
(3) It is not possible to determine the system's input and output matrices b and c with the given information
(4) The input matrix of the system G is b = 4
(5) The output matrix of the system G is c = -1
(6) The input matrix of the system G is b = -1
(7) The output matrix of the system G is c = 2
Award full credit only for unambiguous selection of option (7): The output matrix of the system G is c = 2. Otherwise 0 unless the rubric allows partial credit.

q17 - Consider the output of the parallel system G consisting of G_1 and G_... - 1 point:

Question (q17): Consider the output of the parallel system G consisting of G_1 and G_2. Select the correct statement.
Options:
(1) None of the other statements is correct
(2) The system G is stable if and only if G_1 is unstable and G_2 is stable
(3) The system G is stable if and only if G_1 is stable and G_2 is stable
(4) The system G is stable if and only if G_1 is unstable and G_2 is unstable
Award full credit only for unambiguous selection of option (1): None of the other statements is correct. Otherwise 0 unless the rubric allows partial credit.

q18 - For which z in C does the z-transform of h[n] = a^n - b^n (for n > 0)... - 1 point:

Question (q18): For which z in C does the z-transform of h[n] = a^n - b^n (for n > 0) exist?
Options:
(1) |z| < |a|, |z| < |b|
(2) |z| > |a|, |z| < |b|
(3) |z| < |a|, |z| > |b|
(4) |z| <= |a|, |z| <= |b|
(5) |z| >= |a|, |z| >= |b|
(6) |z| > |a|, |z| > |b|
Award full credit only for unambiguous selection of option (6): |z| > |a|, |z| > |b|. Otherwise 0 unless the rubric allows partial credit.

q19 - For which values of a and b is the output of the system always y[n] = 0? - 1 point:

Question (q19): For which values of a and b is the output of the system always y[n] = 0?
Options:
(1) a > b
(2) a < b
(3) a > b + 1
(4) a = b - 1
(5) a = b
(6) a = f + 1
Award full credit only for unambiguous selection of option (5): a = b. Otherwise 0 unless the rubric allows partial credit.

q20 - Let a = 2b. If input is u[n] = 1^n, find b such that output y[n] = 1... - 1 point:

Question (q20): Let a = 2b. If input is u[n] = 1^n, find b such that output y[n] = 1 for all n.
Options:
(1) b = +/- sqrt(2)
(2) b = 1 +/- sqrt(2)
(3) b = +/- 1
(4) b = +/- sqrt(2)/2
(5) b = 1 +/- sqrt(2)/2
(6) b = +/- 1/2
Award full credit only for unambiguous selection of option (5): b = 1 +/- sqrt(2)/2. Otherwise 0 unless the rubric allows partial credit.

q21 - Select the correct statement regarding cascade system G_3 of G_1 and... - 1 point:

Question (q21): Select the correct statement regarding cascade system G_3 of G_1 and G_2.
Options:
(1) G_3 always has a finite impulse response
(2) None of the other statements is correct
(3) G_3 always has an infinite impulse response
(4) If G_1 and G_2 have a finite impulse response then G_3 has a finite impulse response
Award full credit only for unambiguous selection of option (4): If G_1 and G_2 have a finite impulse response then G_3 has a finite impulse response. Otherwise 0 unless the rubric allows partial credit.

q22 - Assume h_1[n] = 1 for n >= 0 and h_4[0] = 2. Which value corresponds... - 1 point:

Question (q22): Assume h_1[n] = 1 for n >= 0 and h_4[0] = 2. Which value corresponds to h_3[0]?
Options:
(1) h_3[0] = 4
(2) h_3[0] = 3

(3) h_3[0] = 1
(4) It is not possible to determine h_3[0] with the given information
(5) h_3[0] = 2
Award full credit only for unambiguous selection of option (3): h_3[0] = 1. Otherwise 0 unless the rubric allows partial credit.

q23 - Given step responses r_3[n] and r_4[n] and impulse forms for G_1 and... - 1 point:

Question (q23): Given step responses r_3[n] and r_4[n] and impulse forms for G_1 and G_2 with a > b, determine a and b.
Options:
(1) a = 3, b = -1, and c = 0
(2) a = 2, b = 1, and c = -1
(3) a = 5, b = 0, and c = 3
(4) a = 1, b = 0, and c = 1
(5) a = 1, b = 0, and c = -2
(6) a = 3, b = 2, and c = 2
(7) a = 2, b = 1, and c = 4
(8) a = 4, b = -1, and c = 0
Award full credit only for unambiguous selection of option (2): a = 2, b = 1, and c = -1. Otherwise 0 unless the rubric allows partial credit.

q24 - Determine the transfer function W(z) from q[n] to y[n] as a function... - 1 point:

Question (q24): Determine the transfer function W(z) from q[n] to y[n] as a function of a, b, c, d.
Options:
(1) W(z) = (dz + cb - da) / b
(2) W(z) = (bz + c - da) / d
(3) W(z) = (cz - da) / b
(4) W(z) = (dz + cb - da) / (z - a)
(5) W(z) = (dz + cb - da) / z
(6) W(z) = ((d + a)z - cb) / b
Award full credit only for unambiguous selection of option (4): W(z) = (dz + cb - da) / (z - a). Otherwise 0 unless the rubric allows partial credit.

q25 - Determine the values a, b, c, d for the given LTI system state-space... - 1 point:

Question (q25): Determine the values a, b, c, d for the given LTI system state-space representation.
Options:
(1) a = 0.5, b = 0.5, c = 1, d = 1
(2) a = 1, b = 0.5, c = 1, d = 0.5
(3) a = 0.5, b = 3, c = 1, d = 0
(4) a = 0.5, b = 0, c = 1, d = 2
(5) a = 0.8, b = 2, c = 1, d = 0.5
(6) a = 0.5, b = 1, c = 0.5, d = 1
Award full credit only for unambiguous selection of option (1): a = 0.5, b = 0.5, c = 1, d = 1. Otherwise 0 unless the rubric allows partial credit.

q26 - The largest ROC of a transfer function is 0.5 < |z| < 1.5. Determine... - 1 point:

Question (q26): The largest ROC of a transfer function is 0.5 < |z| < 1.5. Determine which pole-zero plot represents H(z).
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (1): Plot A. Otherwise 0 unless the rubric allows partial credit.

q27 - Consider x(t) = sgn(sin(4pi t)). For which sampling time Ts will x(t)... - 1 point:

Question (q27): Consider x(t) = sgn(sin(4pi t)). For which sampling time Ts will x(t) experience aliasing?
Options:
(1) None of the others answers is correct
(2) Ts = 0.01
(3) Ts = 0.1

(4) Ts = 0.05
(5) Ts = 0.2
(6) Ts = 0.5
Award full credit only for unambiguous selection of option (6): Ts = 0.5. Otherwise 0 unless the rubric allows partial credit.

q28 - If the signal is sampled with Ts = 0.001, what is the fundamental per... - 1 point:

Question (q28): If the signal is sampled with Ts = 0.001, what is the fundamental period N?
Options:
(1) N = 10
(2) N = 5000
(3) None of the others answers is correct
(4) N = 1000
(5) N = 50
(6) N = 5
(7) N = 500
(8) N = 100
Award full credit only for unambiguous selection of option (7): N = 500. Otherwise 0 unless the rubric allows partial credit.

q29 - Let {x_1[n]} = {cos(4pi/9 * n)}. Which plot shows the magnitude of {X... - 1 point:

Question (q29): Let {x_1[n]} = {cos(4pi/9 * n)}. Which plot shows the magnitude of {X_1[k]}?
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
(5) Plot E
(6) Plot F
Award full credit only for unambiguous selection of option (4): Plot D. Otherwise 0 unless the rubric allows partial credit.

q30 - Let {x_2[n]} = {0.5 e^(j 3pi n / 9) - 0.5 e^(j 2pi n / 3)}. Which plo... - 1 point:

Question (q30): Let {x_2[n]} = {0.5 e^(j 3pi n / 9) - 0.5 e^(j 2pi n / 3)}. Which plot shows the magnitude of {X_2[k]}?
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
(5) Plot E
(6) Plot F
Award full credit only for unambiguous selection of option (2): Plot B. Otherwise 0 unless the rubric allows partial credit.

q31 - Let {x_3[n]} = {alpha 1^n}. Which plot shows the magnitude of {X_3[k]}? - 1 point:

Question (q31): Let {x_3[n]} = {alpha 1^n}. Which plot shows the magnitude of {X_3[k]}?
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
(5) Plot E
(6) Plot F
Award full credit only for unambiguous selection of option (6): Plot F. Otherwise 0 unless the rubric allows partial credit.

q32 - The DT Fourier transform of output is Y(Omega) = 1 + e^(-j Omega). To... - 1 point:

Question (q32): The DT Fourier transform of output is Y(Omega) = 1 + e^(-j Omega). To what signal does it correspond in time domain?
Options:
(1) y[n] = s[n]
(2) None of the others answers is correct
(3) y[n] = 2 delta[n-2]

(4) y[n] = 2 sin[n]
(5) y[n] = sin[n-1]
(6) y[n] = delta[n] + 1
(7) y[n] = delta[n+2] + delta[n]
(8) y[n] = cos[n+1]
(9) y[n] = 2 cos[n]
(10) y[n] = delta[n] + delta[n-1]
Award full credit only for unambiguous selection of option (10): y[n] = delta[n] + delta[n-1]. Otherwise 0 unless the rubric allows partial credit.

q33 - Consider the magnitude of DT Fourier transform of real sequence {x[n]... - 1 point:

Question (q33): Consider the magnitude of DT Fourier transform of real sequence {x[n]} (given plot). Which statement is correct?
Options:
(1) The sequence {x[n]} is not absolutely summable.
(2) The sequence {x[n]} has a fundamental period of N = 4.
(3) Sum from -inf to inf of x[n] = 4.
(4) None of the others answers is correct
(5) The sequence {x[n]} has a fundamental period of N = 8.
Award full credit only for unambiguous selection of option (1): The sequence {x[n]} is not absolutely summable.. Otherwise 0 unless the rubric allows partial credit.

q34 - Determine the DFS coefficients of u_3[n] = u_1[n] + u_2[n] with given... - 1 point:

Question (q34): Determine the DFS coefficients of u_3[n] = u_1[n] + u_2[n] with given DFS coefficients.
Options:
(1) {U_3[k]} = {0, 2j, 2, -2j}
(2) {U_3[k]} = {2, 2, -2j, 2}
(3) {U_3[k]} = {2, -j, 1, j}
(4) {U_3[k]} = {2, -2j, 2, 2j}
(5) {U_3[k]} = {0, -2j, 2, 2j}
(6) {U_3[k]} = {0, -2j, 4, 2j}
Award full credit only for unambiguous selection of option (6): {U_3[k]} = {0, -2j, 4, 2j}. Otherwise 0 unless the rubric allows partial credit.

q35 - Assume u_3[n] = alpha_1 u_1[n] + alpha_2 u_2[n] and u_3[n] = {1, 2, 1... - 1 point:

Question (q35): Assume u_3[n] = alpha_1 u_1[n] + alpha_2 u_2[n] and u_3[n] = {1, 2, 1, -4}. Determine alpha_1 and alpha_2.
Options:
(1) alpha_1 = 8, alpha_2 = 6
(2) alpha_1 = 1, alpha_2 = 3
(3) alpha_1 = 8, alpha_2 = 3
(4) alpha_1 = 2, alpha_2 = 6
(5) alpha_1 = 2, alpha_2 = 3
(6) alpha_1 = 2, alpha_2 = 4
Award full credit only for unambiguous selection of option (2): alpha_1 = 1, alpha_2 = 3. Otherwise 0 unless the rubric allows partial credit.

q36 - Determine the DFS coefficients of y_3[n] = G(u_1[n] + u_2[n]). - 1 point:

Question (q36): Determine the DFS coefficients of y_3[n] = G(u_1[n] + u_2[n]).
Options:
(1) {Y_3[k]} = {0, -2j, 8, 2j}
(2) {Y_3[k]} = {0, 0, 8, 0}
(3) {Y_3[k]} = {0, 0, 16, 0}
(4) {Y_3[k]} = {2, 2j, 8, 2j}
(5) {Y_3[k]} = {2, 4j, 0, -4j}
(6) {Y_3[k]} = {2, -4j, 0, 4j}
Award full credit only for unambiguous selection of option (2): {Y_3[k]} = {0, 0, 8, 0}. Otherwise 0 unless the rubric allows partial credit.

q37 - Which of the following plots represents the frequency response H(Omega)? - 1 point:

Question (q37): Which of the following plots represents the frequency response H(Omega)?
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (3): Plot C. Otherwise 0 unless the rubric allows partial credit.

q38 - If input signal is u[n] = 2 + cos(pi/2 * n), what corresponds to outp... - 1 point:

Question (q38): If input signal is u[n] = 2 + cos(pi/2 * n), what corresponds to output signal y[n]?
Options:
(1) y[n] = 2 + cos(pi/2 * n - pi/2)
(2) y[n] = 0
(3) y[n] = 2
(4) y[n] = 4
(5) y[n] = 2 + sin(pi/2 * n - pi/2)
(6) y[n] = 1 + cos(pi/2 * n)
Award full credit only for unambiguous selection of option (3): y[n] = 2. Otherwise 0 unless the rubric allows partial credit.

q39 - Determine the frequency response H_2(Omega) relative to system G_2. - 1 point:

Question (q39): Determine the frequency response H_2(Omega) relative to system G_2.
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (2): Plot B. Otherwise 0 unless the rubric allows partial credit.

q40 - Assume U(z) = 1 - 0.25 z^(-1). Determine the frequency response H_3(O... - 1 point:

Question (q40): Assume U(z) = 1 - 0.25 z^(-1). Determine the frequency response H_3(Omega) relative to system G_3.
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (3): Plot C. Otherwise 0 unless the rubric allows partial credit.

q41 - Let y_1[n] = |x_1[n]| + |x_2[n]| and y_2[n] = |x_1[n] + x_2[n]|. Whic... - 1 point:

Question (q41): Let y_1[n] = |x_1[n]| + |x_2[n]| and y_2[n] = |x_1[n] + x_2[n]|. Which statement is true for Y_1[0] and Y_2[0]?
Options:
(1) Y_1[0] >= Y_2[0]
(2) Y_1[0] = Y_2[0]
(3) Y_1[0] <= Y_2[0]
(4) None of the other statements can be determined.
(5) Y_1[0] < Y_2[0]
(6) Y_1[0] > Y_2[0]
Award full credit only for unambiguous selection of option (1): Y_1[0] >= Y_2[0]. Otherwise 0 unless the rubric allows partial credit.

q42 - Assume x_1[n] = (1)^n and x_2[n] = cos(pi n) for length N=2. Which st... - 1 point:

Question (q42): Assume x_1[n] = (1)^n and x_2[n] = cos(pi n) for length N=2. Which statement is true regarding sum of DFT coefficients?
Options:
(1) Sum Y_1 < Sum Y_2
(2) Sum Y_1 <= Sum Y_2
(3) Sum Y_1 >= Sum Y_2
(4) Sum Y_1 = Sum Y_2

(5) Sum Y_1 > Sum Y_2
(6) None of the other statements can be determined.
Award full credit only for unambiguous selection of option (4): Sum Y_1 = Sum Y_2. Otherwise 0 unless the rubric allows partial credit.

q43 - Which of the following filters G_A, G_B, G_C, G_D is a low-pass filter? - 1 point:

Question (q43): Which of the following filters G_A, G_B, G_C, G_D is a low-pass filter?
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) None of the others
Award full credit only for unambiguous selection of option (1): G_A. Otherwise 0 unless the rubric allows partial credit.

q44 - Which of the following filters G_A, G_B, G_C, G_D is a high-pass filter? - 1 point:

Question (q44): Which of the following filters G_A, G_B, G_C, G_D is a high-pass filter?
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) None of the others
Award full credit only for unambiguous selection of option (3): G_C. Otherwise 0 unless the rubric allows partial credit.

q45 - Which of the following filters G_A, G_B, G_C, G_D is a band-pass filter? - 1 point:

Question (q45): Which of the following filters G_A, G_B, G_C, G_D is a band-pass filter?
Options:
(1) G_A
(2) G_B
(3) G_C
(4) G_D
(5) None of the others
Award full credit only for unambiguous selection of option (2): G_B. Otherwise 0 unless the rubric allows partial credit.

q46 - What kind of filter does the pole-zero plot in Box 16 implement? - 1 point:

Question (q46): What kind of filter does the pole-zero plot in Box 16 implement?
Options:
(1) Notch filter
(2) Band-stop filter
(3) None of these
(4) Band-pass filter
(5) High-pass filter
(6) Low-pass filter
Award full credit only for unambiguous selection of option (5): High-pass filter. Otherwise 0 unless the rubric allows partial credit.

q47 - If the depicted frequency transformation is applied to a low-pass fil... - 1 point:

Question (q47): If the depicted frequency transformation is applied to a low-pass filter, what kind of filter is obtained?
Options:
(1) Band-stop filter
(2) Low-pass filter
(3) Band-pass filter
(4) High-pass filter
(5) Notch filter
(6) None of these
Award full credit only for unambiguous selection of option (1): Band-stop filter. Otherwise 0 unless the rubric allows partial credit.

q48 - Which DFT coefficient would you set to zero in order to attenuate a d... - 1 point:

Question (q48): Which DFT coefficient would you set to zero in order to attenuate a disturbance at 100Hz?
Options:
(1) k = 10 and k = 31
(2) k = 10 and k = 91
(3) k = 50
(4) k = 40
(5) k = 5 and k = 96
(6) k = 5 and k = 11
Award full credit only for unambiguous selection of option (2): k = 10 and k = 91. Otherwise 0 unless the rubric allows partial credit.

q49 - Which DFT coefficients would you set to zero in order to filter all c... - 1 point:

Question (q49): Which DFT coefficients would you set to zero in order to filter all components higher than 200Hz?
Options:
(1) k <= 20
(2) k >= 30
(3) k >= 30 and k <= 71
(4) k <= 40
(5) k >= 40
(6) k >= 20 and k <= 81
Award full credit only for unambiguous selection of option (6): k >= 20 and k <= 81. Otherwise 0 unless the rubric allows partial credit.

q50 - Consider sampled sine signal. Mark the correct statement regarding al... - 1 point:

Question (q50): Consider sampled sine signal. Mark the correct statement regarding aliasing during up/down sampling.
Options:
(1) Even if x[n] didn't experience aliasing then up-sampling x[n] can cause aliasing
(2) If f in N and Ts in N then x[n] will not be periodic
(3) If f = 30 then Ts >= 0.3 will avoid aliasing
(4) Even if x[n] didn't experience aliasing then down-sampling x[n] can cause aliasing
(5) None of the others
Award full credit only for unambiguous selection of option (4): Even if x[n] didn't experience aliasing then down-sampling x[n] can cause aliasing. Otherwise 0 unless the rubric allows partial credit.

q51 - Determine DFS coefficients Y_1[k] of output after applying moving ave... - 1 point:

Question (q51): Determine DFS coefficients Y_1[k] of output after applying moving average filter to u_1[n].
Options:
(1) Y_1[k] = {3, 0, 0}
(2) Y_1[k] = {0, 1.5, 1.5}
(3) Y_1[k] = {0, 0, 0}
(4) Y_1[k] = {0, 1.5j, -1.5j}
(5) Y_1[k] = {3, 0, 1.5}
(6) None of the others
Award full credit only for unambiguous selection of option (1): Y_1[k] = {3, 0, 0}. Otherwise 0 unless the rubric allows partial credit.

q52 - Determine output y_2[n] after applying filter h[n] to input u_2[n]. - 1 point:

Question (q52): Determine output y_2[n] after applying filter h[n] to input u_2[n].
Options:
(1) y_2[n] = alpha
(2) y_2[n] = 1
(3) y_2[n] = 1 + alpha cos(pi/5)
(4) y_2[n] = 1 + alpha cos(2pi/5 + pi/5)
(5) y_2[n] = 1 + alpha cos(2pi/5)
(6) y_2[n] = beta + alpha sin(pi/5)
Award full credit only for unambiguous selection of option (5): y_2[n] = 1 + alpha cos(2pi/5). Otherwise 0 unless the rubric allows partial credit.

q53 - Determine output y_1[n] after applying filter h[n] to input u_1[n] wi... - 1 point:

Question (q53): Determine output y_1[n] after applying filter h[n] to input u_1[n] with order M=5.
Options:
(1) y_2[n] = 1
(2) y_2[n] = 1 + alpha cos(pi/3 + pi)
(3) y_2[n] = 1 + alpha cos(2pi/3 + pi)
(4) y_2[n] = 1 + alpha sin(2pi/3)
(5) y_2[n] = 1 + beta sin(2pi/3 + pi)
(6) y_2[n] = 1 + beta cos(pi/3)
Award full credit only for unambiguous selection of option (3): y_2[n] = 1 + alpha cos(2pi/3 + pi). Otherwise 0 unless the rubric allows partial credit.

q54 - Determine w_c such that z_p = 0.5 for the discretized high-pass filter. - 1 point:

Question (q54): Determine w_c such that z_p = 0.5 for the discretized high-pass filter.
Options:
(1) w_c = 1
(2) w_c = 2
(3) w_c = 8
(4) w_c = 6
(5) w_c = 10
(6) w_c = 3
Award full credit only for unambiguous selection of option (4): w_c = 6. Otherwise 0 unless the rubric allows partial credit.

q55 - Determine frequency transformation f(z) for the band-pass filter H_bp... - 1 point:

Question (q55): Determine frequency transformation f(z) for the band-pass filter H_bp(z) = H_hp(f(z)).
Options:
(1) f(z) = -z
(2) f(z) = z
(3) f(z) = z^3
(4) f(z) = z^2
(5) f(z) = -z^3
(6) f(z) = -z^2
Award full credit only for unambiguous selection of option (4): f(z) = z^2. Otherwise 0 unless the rubric allows partial credit.

q56 - Determine desired cut-off frequencies Omega_0^des and Omega_1^des as... - 1 point:

Question (q56): Determine desired cut-off frequencies Omega_0^des and Omega_1^des as functions of z_p.
Options:
(1) Omega_0^des = (1 + z_p) / (1 - z_p) and Omega_1^des = (1 + z_p) / (1 - z_p) + pi
(2) None of the others
(3) Omega_0^des = arctan z_p and Omega_1^des = arctan z_p + pi
(4) Omega_0^des = 1 + z_p and Omega_1^des = 1 - z_p + pi
(5) Omega_0^des = arctan((1 + z_p) / (1 - z_p)) and Omega_1^des = arctan((1 + z_p) / (1 - z_p)) + pi
Award full credit only for unambiguous selection of option (5): Omega_0^des = arctan((1 + z_p) / (1 - z_p)) and Omega_1^des = arctan((1 + z_p) / (1 - z_p)) + pi. Otherwise 0 unless the rubric allows partial credit.

q57 - Which frequency mapping f(s) would you choose for generating the band... - 1 point:

Question (q57): Which frequency mapping f(s) would you choose for generating the band-stop filter from a second-order low-pass Butterworth filter?
Options:
(1) f(s) = s^2 / (s^2 + w_0 w_1)
(2) None of the others
(3) f(s) = (s^2 + w_0 w_1) / s
(4) f(s) = s / (s^2 + w_0 w_1)
(5) f(s) = s / (s^2 + (w_0 w_1)^2)
Award full credit only for unambiguous selection of option (4): f(s) = s / (s^2 + w_0 w_1). Otherwise 0 unless the rubric allows partial credit.

q58 - Which second-order low-pass Butterworth filter could have been used t... - 1 point:

Question (q58): Which second-order low-pass Butterworth filter could have been used to generate the band-stop filter with w_0=5 and w_1=10?
Options:
(1) H_LP(s) = 0.04 / (s^2 + 0.2828s + 0.04)
(2) H_LP(s) = 4 / (s^2 + 2.828s + 4)
(3) H_LP(s) = 0.01 / (s^2 + 0.1414s + 0.01)
(4) H_LP(s) = 1 / (s^2 + 1.414s + 1)
(5) H_LP(s) = 25 / (s^2 + 7.071s + 25)
Award full credit only for unambiguous selection of option (1): H_LP(s) = 0.04 / (s^2 + 0.2828s + 0.04). Otherwise 0 unless the rubric allows partial credit.

q59 - Consider H(z) = 1 / (z(z - a)). Input u[n] = sin(pi/2 * n) yields out... - 1 point:

Question (q59): Consider H(z) = 1 / (z(z - a)). Input u[n] = sin(pi/2 * n) yields output with phase atan(-2). What is a?
Options:
(1) a = 2
(2) a = 0.1
(3) It is not possible to identify the system with this information
(4) a = 1
(5) None of the others
(6) a = 0.5
Award full credit only for unambiguous selection of option (1): a = 2. Otherwise 0 unless the rubric allows partial credit.

q60 - What is the amplitude A of the applied sinusoidal input in the experi... - 1 point:

Question (q60): What is the amplitude A of the applied sinusoidal input in the experimental frequency response identification procedure?
Options:
(1) A = 0.5
(2) A = 0.1
(3) A = 5
(4) A = 0.2
(5) A = 2
(6) A = 10
Award full credit only for unambiguous selection of option (2): A = 0.1. Otherwise 0 unless the rubric allows partial credit.